

Domain, Interpretation, and Semantics

Video Lesson

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Slide Deck

M09S02 (<https://canvas.tamu.edu/courses/430127/files/92570674?wrap=1>) 

Lesson Transcript

In this lecture, we will look into domain interpretation and semantics. We will need these concepts to determine the truth value of a particular formula in first-order logic.

Domain

- A *Domain* is a section of the world about which we wish to express some knowledge.
- The totality of the objects in the part of that world consists the **domain**.
- Basically, the set of all **constants** (i.e. objects) makes up a domain: *John, Bill, Bob, ...*

Example: everyone on Earth, everyone in Texas, everyone in College Station, every computer in the Petersen Bldg., etc.

A domain is a section of the world about which we wish to express some knowledge. The totality of the objects in that part of the world consists of the domain.

Basically, the set of all constants, that is, the objects, makes up a domain. For example, John, Bill, and Bob, in case it is about the population.

Other examples include everyone on Earth, everyone in Texas, everyone in College Station, every computer in the Peterson building, and so on, depending on what we want to model.

Example: Kinship Domain

$$\forall m \forall c \text{ Mother}(c) = m \leftrightarrow (\text{Female}(m) \wedge \text{Parent}(m, c))$$
$$\forall w \forall h \text{ Husband}(h, w) \leftrightarrow \text{Male}(h) \wedge \text{Spouse}(h, w)$$
$$\forall p \forall c \text{ Parent}(p, c) \leftrightarrow \text{Child}(c, p)$$
$$\forall g \forall c \text{ Grandparent}(g, c) \leftrightarrow \\ \exists p (\text{Parent}(g, p) \wedge \text{Parent}(p, c))$$
$$\forall x \forall y \text{ Sibling}(x, y) \leftrightarrow \\ (x \neq y \wedge \exists p (\text{Parent}(p, x) \wedge \text{Parent}(p, y)))$$

Exercise: 7.6, p. 214 – try it out

We can also have some rules to describe the relationship between these objects in the domain. So, for example, here is the kinship domain where family relations are indicated. For all m , for all c , if $\text{Mother}(c)$ equals m , then if and only if m is a female and m is a parent of c .

Next is for all w , and for all h , if h is the husband of w , then if and only if h is male and h is a spouse of w . So, here we are establishing the relationship between this husband predicate and

male and spouse predicates. Likewise, in the first sentence, we are trying to establish the relationship between the female and parent predicates and the mother predicate, and the rest is pretty self-explanatory.

Parent and child relationship. So, as we can see here, the ordering is different. p is a parent of the child, and c is a child of p . So, here the order is p and c , but for the child predicate, the order is c and p . And grandparent and relationship with the parent predicate, and sibling, and its relationship with the parent predicate.

So, of course, here, we have to exclude the case where x and y are identical. The reason is we are not a sibling of ourselves.

Interpretation in Predicate Calculus

An **interpretation** of a formula F in first-order logic consists of a nonempty domain D , and an assignment of values to each constant, function, and predicate occurring in F as follows:

- ① **constant**:
assign an element of D (e.g. an integer)
- ② **function** with n arguments:
assign a mapping from D^n to D
- ③ **predicate** with n arguments:
assign a mapping from D^n to $\{\mathbf{True}, \mathbf{False}\}$

$D^n = \{(x_1, \dots, x_n) | x_i \in D \text{ for } i = 1, \dots, n\}$ Similar to assigning truth values in propositional logic.

Interpretation in propositional logic was pretty simple. We assigned the **True** or *False* value to each of the atoms, but in predicate calculus or first-order logic, we have to specify all of these things, the constants, the functions, and the predicates.

So, for each constant, we assign an element of the domain D , and for functions, we have to provide all of these mappings. For a function with n arguments, the mapping is from an n -dimensional space of values, D , to a 1-dimensional space in the domain.

Predicates should also have defined values, **True** or **False**, for the particular arguments.

Evaluation of a Formula Given an Interpretation



- 1 if F and G have been evaluated, evaluate $\neg F$, $(F \wedge G)$, $(F \vee G)$, $(F \rightarrow G)$, and $(F \leftrightarrow G)$
- 2 $\forall x G$ is evaluated to **True** if G evaluates to **True** for every x in D ; otherwise, G evaluates to **False**
- 3 $\exists x G$ is evaluated to **True** if G evaluates to **True** for any x in D ; otherwise, G evaluates to **False**

Given an arbitrary formula in first-order logic, if F and G have been evaluated already, then we can trivially evaluate these expressions that include the logical connectives or negation. So, if F is evaluated as **True**, then $\neg F$ would be **False**, and so on.

Because we have quantifiers in first-order logic, we have to treat them carefully. So, $\forall x G$ is evaluated to **True** if this G evaluates to **True** for every x in the domain D . Otherwise, G evaluates to **False**.

For the existential quantifier, $\exists x G$ is evaluated to **True** if G evaluates to **True** for any x in D . That is, if we can find at least one example that satisfies G , then this expression is **True**, and otherwise, it is **False**.

Side Note: Bound vs. Free Variables

- **Scope of a quantifier:**
the range (parentheses) over which the associated variable takes effect
- **Bound variable:**
an occurrence of a variable in a formula is **bound** iff the occurrence is within the scope of a quantifier employing the variable.
- **Free variable:**
an occurrence of a variable in a formula is **free** iff the occurrence is not bound.

Bound: $\forall x \forall y P(x, y)$; Free: $\forall x P(x, y)$;

Both Free and Bound: $(\forall x P(x, y)) \wedge (\forall y Q(y))$

The scope of a quantifier is the range over which the associated variables take effect. This is typically indicated by the parentheses.

And there could be two types of variables. One is a bound variable, and the other is a free variable. A bound variable is always associated with one of these two different quantifiers, the universal quantifier or the extension quantifier.

So, here is an example: x and y are bound variables, whereas in this case, y is standalone, and there is no associated quantifier. So, in this case, y is a free variable, whereas x is a bound variable.

In our course, we will only consider bound variables. So, this is just an interesting concept, but we do not need to pay too much attention to it.

Example: Interpretation and Evaluation

Given the **interpretation**:

- **Domain:** $D = \{Bob, Carol, Ted, Alice\}$
- **Predicates:** $Woman(Carol), Woman(Alice)$
 $Man(Bob), Man(Ted)$
 $Loves(Bob, Carol), Loves(Ted, Alice), Loves(Carol, Ted)$
- **Functions:** $Brother(Bob) = Ted, Boss(Alice) = Carol$

Evaluate:

$$\forall x (Man(x) \rightarrow \exists y (Woman(y) \wedge Loves(x, y)))$$

As mentioned earlier in propositional logic, interpretation is simply a truth value assigned to the atoms. However, in first-order logic, we have to fully specify what the domain is, what the predicate values are, the **True** or **False** values, and what the function values are.

So, in considering whether this expression is **True** or **False**, we must have this full interpretation.

Because the domain D includes only four people, it means that our entire population consists of only four people.

So, when we see $\forall x$, this x is any one of these, and if this particular sentence here, $Man(x)$ implies there exists y where y is a woman, and x loves y . To tell whether this sentence is **True** or **False**, we have to plug *Bob* into x , *Carol* into x , *Ted* into x , and *Alice* into x , and see whether this sentence is **True**, and only when for all of these four individuals this sentence is **true**, then this entire expression is **True**.

Example: Interpretation and Evaluation

Given the interpretation:

- **Domain:** $D = \{Bob, Carol, Ted, Alice\}$
- **Predicates:** $Woman(Carol), Woman(Alice)$
 $Man(Bob), Man(Ted)$
 $Loves(Bob, Carol), Loves(Ted, Alice), Loves(Carol, Ted)$
- **Functions:** $Brother(Bob) = Ted, Boss(Alice) = Carol$

Evaluate:

$$\forall x (Man(x) \rightarrow \exists y (Woman(y) \wedge Loves(x, y)))$$

① $x = Bob$

$$\frac{Man(Bob)}{True} \rightarrow \frac{\exists y (Woman(y) \wedge Loves(Bob, y))}{True \text{ when } y = Carol}$$

Let us look at the first case where x is *Bob*. Then we plug *Bob* into x , here and here, and we get this sentence, $Man(Bob)$ implies there exists y such that y is a woman and *Bob* loves this woman, y . First, this is true because it is given here in our interpretation. *Bob* is a man. And this is also true when y is *Carol*.

So, if we look at this *Loves* predicate, *Bob* loves *Carol*, and it happens that *Carol* is also a woman. So, these three predicates, when we plug them in here, will satisfy that formula.

In the second case, this plugin *Carol* for the variable x , then this sentence becomes $Man(Carol)$ implies something here. However, because $Man(Carol)$ is not in the predicates with a **True** value, this is **False**.

Because the premise part of this implication is **False**, this entire sentence becomes **True**.

So far, we have shown that this sentence is **True**. When x is *Bob* or x is *Carol*.



Example: Interpretation and Evaluation

Given the **interpretation**:

- **Domain:** $D = \{Bob, Carol, Ted, Alice\}$
- **Predicates:** $Woman(Carol), Woman(Alice)$
 $Man(Bob), Man(Ted)$
 $Loves(Bob, Carol), Loves(Ted, Alice), Loves(Carol, Ted)$
- **Functions:** $Brother(Bob) = Ted, Boss(Alice) = Carol$

Evaluate:

$$\forall x (Man(x) \rightarrow \exists y (Woman(y) \wedge Loves(x, y)))$$

$$\begin{array}{l} \textcircled{1} x = Bob \\ \frac{Man(Bob) \rightarrow \exists y (Woman(y) \wedge Loves(Bob, y))}{True} \quad \text{True when } y = Carol \end{array}$$

$$\begin{array}{l} \textcircled{2} x = Carol \\ \frac{Man(Carol) \rightarrow \dots}{False} \quad \text{So, this is True} \end{array}$$

$$\textcircled{3} x = Ted$$

$$\textcircled{4} x = Alice$$

The rest is very similar. So, when x is *Ted*, that x equals *Ted* is a man, and *Ted* loves *Alice*, and *Alice* is a woman. So, this sentence is satisfied when x equals *Ted*. So, this is also **True**. Similar to $x = Carol$, when x is *Alice*, because *Alice* is not a man, this expression will turn out to be **True**. So, for all of these four people in the domain, this sentence turns out to be **True** given these truth assignments.

So, this sentence as a whole is evaluated to **>True**.

Example: Interpretation and Evaluation

$$\forall x (Man(x) \rightarrow \exists y (Woman(y) \wedge Loves(x, y)))$$

- Must consider $x \in \{Bob, Carol, Ted, Alice\}$.
- $x = Carol$ or $x = Alice$:
since $\neg Man(Carol)$ and $\neg Man(Alice)$, formula is true.
- $x = Bob$:
Since $Loves(Bob, Carol)$ and $Woman(Carol)$, formula is true.
- $x = Ted$:
Since $Loves(Ted, Alice)$ and $Woman(Alice)$, formula is true.

Here is a summary of what we saw in the previous slide.

So again, to repeat, we have to consider x in the domain for all four of these people. So, when x is *Carol* or x is *Alice*, because they are not male, the formula is **True**. Again, because the premise part is **False**, and for *Bob* and *Ted*, because they love someone who happens to be a woman, for the case of *Bob*, y would be *Carol*. In the case of *Ted*, y would be *Alice*, and these would satisfy the sentence.

Consistency, Satisfiability, and Validity

A formula G is

- **consistent** (satisfiable) iff there exists an interpretation I such that G evaluates to **True** in I . In this case, I is a **model** of G and I satisfies G .
- **inconsistent** (unsatisfiable) iff there is no interpretation that satisfies G .
- **valid** iff every interpretation of G satisfies G .
- **invalid** iff there is at least one interpretation I of G such that G evaluates to **False** under I .

These first-order logic formulas can have different properties depending on which conditions are **True** or **False**. These are consistent, inconsistent, valid, or invalid. A formula G is consistent, otherwise called satisfiable, if and only if there exists an interpretation I such that G evaluates to **True** under this interpretation I . So, if we can find just one example of that, then it is satisfiable.

In this case, I is called a model of G , and also it is said that I satisfies G .

A formula G is inconsistent or unsatisfiable if and only if there is no interpretation that satisfies this G . A formula G is valid if and only if every interpretation of G satisfies G . And finally, a formula G is invalid if and only if there is at least one interpretation I of G such that this G evaluates to **False** under this particular interpretation, I .

Difficulty: Many Domains Are Infinite

Algebra, etc.

- There are an infinite number of interpretations of a formula.
- In general, **none** of the properties in the previous slide are **decidable** in general for formulas in predicate calculus.

One of the main difficulties here is that many domains are infinite, for example, algebra.

There are an infinite number of interpretations for a formula. And in general, none of these properties we saw in the previous slide can be decidable.

So far, we looked at the basics of first-order logic and how to assign the truth value for an arbitrary sentence or formula in first-order logic.

In the following lectures, we will look at the standard or canonical form, which is the basic requirement on our journey to fully automated theorem proving in first-order logic.